

Suspicious PowerShell Activity Investigation Report

Executive Summary

This report documents the investigation of suspicious PowerShell activity identified through Splunk log analysis. The objective was to detect potentially risky PowerShell execution patterns including encoded commands, execution policy bypass attempts, hidden execution, and outbound web requests.

The investigation revealed several suspicious activities that align with commonly observed attacker techniques and post-exploitation behavior.

Incident Details

Incident Type

Suspicious PowerShell Activity

Severity

Medium

Detection Method

Splunk Log Analysis

Data Source

PowerShell Execution Logs

Objective

The purpose of this investigation was to:

- Detect encoded PowerShell commands
- Identify execution policy bypass attempts
- Detect hidden PowerShell execution
- Identify suspicious web request activity
- Perform threat hunting
- Develop monitoring and alerting logic

Investigation Methodology

Step 1: Log Collection

PowerShell execution logs were uploaded into Splunk and indexed for analysis.

Step 2: Encoded Command Analysis

Encoded PowerShell commands were searched to identify potential obfuscation attempts.

Query:

```
index=powershell commandline="*-enc*"
```

Result

Multiple encoded command executions were identified.

Step 3: Execution Policy Bypass Detection

Execution policy bypass attempts were analyzed because attackers frequently disable PowerShell restrictions.

Query:

```
index=powershell commandline="*ExecutionPolicy Bypass*"
```

Result

Several policy bypass executions were observed.

Step 4: Web Request Investigation

Outbound web requests were reviewed to identify possible payload downloads.

Query:

```
index=powershell commandline="*Invoke-WebRequest*"
```

Result

PowerShell commands communicating with external domains were detected.

Step 5: Hidden Execution Analysis

Hidden PowerShell windows were investigated because attackers often use them to avoid visibility.

Query:

index=powershell commandline="*Hidden*

Result

Multiple hidden execution events were identified.

Step 6: Threat Hunting

A comprehensive hunting query was executed to consolidate suspicious PowerShell behaviors.

Result

Several events matched patterns associated with attacker tradecraft.

Findings

The investigation identified:

- Encoded PowerShell execution
- Execution Policy Bypass attempts
- Hidden PowerShell activity
- Outbound web requests
- Suspicious command-line behavior

These activities may indicate unauthorized script execution, malware activity, or post-exploitation techniques.

MITRE ATT&CK Mapping

Technique	MITRE ID
PowerShell	T1059.001
Obfuscated Files and Information	T1027
Command and Scripting Interpreter	T1059
Impair Defenses	T1562

Risk Assessment

The observed activity could be associated with:

- Malware execution
- Script-based attacks
- Payload delivery attempts
- Defense evasion techniques
- Unauthorized administrative activity

Further endpoint investigation is recommended.

Recommendations

1. Enable PowerShell logging and auditing.
2. Monitor encoded command execution.
3. Alert on Execution Policy Bypass attempts.
4. Investigate outbound PowerShell web requests.
5. Deploy Sysmon for enhanced visibility.
6. Implement PowerShell security controls.

Conclusion

The investigation successfully identified multiple suspicious PowerShell behaviors through Splunk analysis. Continuous monitoring and detection development can help organizations identify and respond to PowerShell-based threats more effectively.

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